

Oasis Intelligence

B2B AGRI-FOOD SUPPLY CHAIN & FARMLAND INFRASTRUCTURE ANALYTICS PLATFORM

1. Project Title & Technical Classification

Project Title: Oasis Intelligence: A B2B Agri-Food Supply Chain and Farmland Infrastructure Analytics Platform

Core Classification: Enterprise Market Intelligence, Cross-Border Supply Chain Management, and Agronomic Asset Analytics.

2. Project Summary

Oasis Intelligence is an enterprise-level B2B market intelligence and supply chain analytics web application engineered to resolve a critical information asymmetry within the Middle Eastern agricultural sector. While Gulf Cooperation Council (GCC) countries face extreme arid climates with limited arable land, their sovereign-backed mega-conglomerates such as SALIC, Al Dahra, Hassad Food, ARASCO, NADEC, and Baladna invest billions of dollars annually into sophisticated domestic high-tech infrastructure and multi-origin cross-border land acquisitions. However, the operational, logistical, and structural corporate data tracking these massive projects remains highly fragmented across disconnected international ministries, dense corporate disclosures, trade-association listings, and localized commercial registries. Our platform closes this critical operational gap by aggregating, standardizing, and synthesizing these disparate data streams into a unified, interactive intelligence workspace.

The platform maps out comprehensive parent-subsidary enterprise networks, explicit corporate land-bank metrics (differentiating between hectares owned versus leased), infrastructure processing capabilities (such as localized grain storage capacities in metric tons), and complex technology stacks (including industrial irrigation setups and heavy machinery vendor specifications) for the primary controllers of regional food security. Serving as a centralized intelligence system, Oasis Intelligence enables agribusiness executives, global commodities traders, trade finance institutions, and heavy equipment manufacturers to systematically evaluate and navigate the Middle Eastern agrifood landscape. By deploying a robust relational database architecture, the platform empowers users to run multi-parameter search workflows to filter corporate infrastructure assets by geographic coordinates, target crop mix (e.g., wheat, barley, soy, corn), and capacity benchmarks. Ultimately, the system transforms raw public records and asymmetric financial data into strategic corporate maps, providing international stakeholders with the predictive clarity and logistical transparency required to optimize business development, target sales pipelines, and manage cross-border supply chain risks.

3. Creative Component

The creative component of Oasis Intelligence is an interactive data visualization dashboard that displays agricultural trade and food production trends for GCC countries using dynamic charts and plots rendered in the browser. The dashboard includes several interactive graphs such as bar charts comparing import volumes across countries, line graphs showing trade trends over time, and scatter plots comparing domestic production against import dependency. Users can interact with these visualizations by selecting

different countries, commodities, and year ranges from dropdown menus, with the charts updating dynamically based on live queries against the database.

The technical complexity of this component comes from establishing an engineered data pipeline that connects a Visualization library to the custom backend API layer. Because the application strictly operates without an Object Relational Mapper (ORM), the system must handle the processing and parsing of raw, multi-table aggregated SQL results into highly optimized, chart-ready JSON formats on the fly. Furthermore, the application must manage dynamic user input configurations that capture multi-variable filter states to trigger entirely different underlying backend raw queries depending on the selected client-side dropdown parameters.

4. Usefulness & Market Differentiation

Oasis Intelligence brings together agricultural trade and food production data for Gulf Cooperation Council (GCC) countries into a single, searchable web application. GCC countries import the vast majority of their food supply due to limited arable land and arid climates, making this data highly relevant to agribusiness professionals. Currently, this data lives across multiple international databases like FAOSTAT and the World Bank with no unified interface to query and compare it easily. Our application solves this by providing a clean, simple frontend over a well-structured relational database.

- **Basic Functions:** Searching for a country or commodity, browsing overall commodity types, and viewing standalone country profiles that display top imported commodities alongside baseline agricultural production figures.
- **Complex Features:** The platform computes a localized import dependency metric per country, drives an interactive visualization dashboard via dynamic asynchronous lookups, and deploys an atomic bulk data import transaction engine that strictly validates and loads new agricultural records without data duplication.

Similar Applications & Differentiation

Similar tools exist but serve entirely different purposes. FAOSTAT and the World Bank data portal provide raw data downloads but lack a unified search or comparative cross-reference interface. UN Comtrade covers global trade flows but is overly complex to navigate and fails to focus on the unique parameters of the GCC region. Our application stands apart because it combines raw production and macroeconomic trade data from two distinct international sources into a single queryable interface, scopes the data domain specifically to GCC countries, and surfaces a computed dependency metric that none of the existing public tools provide directly to stakeholders.

5. Realness & Data Sources

Our application utilizes two public datasets which provide real, large-scale agriculture and economic data relevant to the middle eastern countries.

Data Source Registry	Data Format	Cardinality (Rows)	Degree	Target Information & Attributes Captured
Dataset 1: FAOSTAT (United Nations Food & Agriculture Organization) Source: https://www.fao.org/fao-stat/en/#data	Structured CSV	~18,000 Production rows; ~4,000 Land Use rows	~12 Cols	Captures Country Name, Country Code, Crop Item, Item Code, Element metrics (Production Vol, Yield, Harvested Area), Unit flags, Year, and Value fields. Traces specific crops (wheat, barley, soy, corn, dates, alfalfa).
Dataset 2: World Bank Open Data (Agriculture & Rural Development) Source: https://data.worldbank.org/topic/agriculture-and-rural-development	Structured CSV / XLS	~6,000 Records	~8 Cols	Captures Country Name, Country Code, Indicator Name (e.g., Food Import Dependency, Irrigated Land %, Ag GDP Contribution), Indicator Code, Fiscal Year, and Value fields.

6. Detailed Functional Specifications (CRUD)

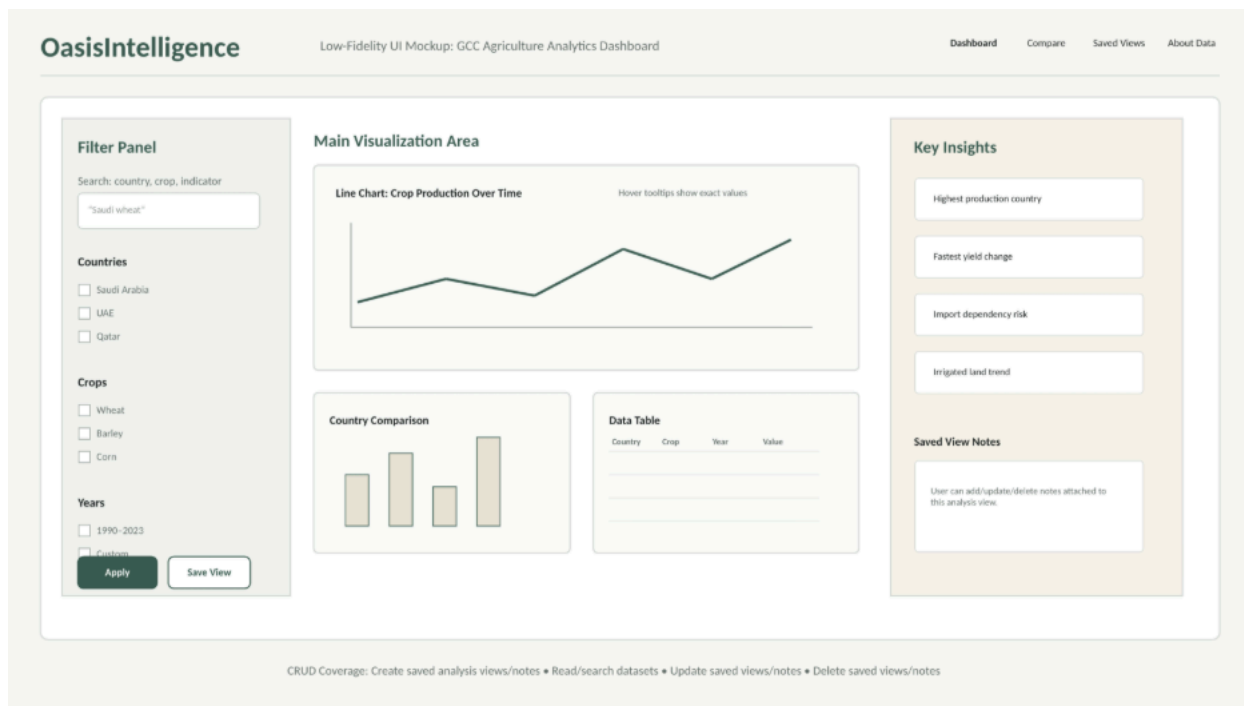
Our application allows users to explore Middle Eastern and GCC agricultural and food-security records through a searchable analytics dashboard. FAOSTAT and World Bank tables are imported as read-only source data, while users are given full CRUD capability to create, read, update, and delete their own saved analysis views and text notes inside an isolated workspace table layer.

System User Interaction Pipeline

- 1. Search Records (Read/Search):** A user submits a search input form with keywords like 'Saudi Arabia wheat production'. The system executes indexed string queries against the database and returns matching records in a paginated list view.
- 2. Filter Dashboard Data (Read/Filter):** A user submits a filter form containing selections for country, crop type, year range, production volume thresholds, or macroeconomic indicator values. The system queries the backend, updating layout visualizations.
- 3. Compare Countries and Crops (Read/Compare):** A user selects multiple countries or crops from an interface menu when they want to compare agricultural capacities across the region. The system fetches the corresponding records and renders side-by-side matrices.
- 4. View Visualizations (Read):** A user interacts with visual components to understand trends more clearly than through raw data grids. The platform displays line charts for production changes, bar charts, and ranking metrics.

- **5. Create Saved Analysis Views (CREATE):** A user saves an active dashboard state layout for later evaluation. They submit a form containing a custom view title, descriptive paragraph, active crop mixes, and optional notes, triggering a raw SQL INSERT query.
- **6. Read/Open Saved Views (READ):** A user navigates to their personal analyst profile workspace. The system runs a SELECT statement to query all configurations tied to their account ID and displays each saved view's filters, timestamps, and notes.
- **7. Update Saved Views and Notes (UPDATE):** A user revises an existing analysis setup inside their profile screen. They alter text notes or adjust active filters, executing a direct raw SQL UPDATE statement matching the specific view tracking key.
- **8. Delete Saved Views (DELETE):** A user purges an obsolete view from their workspace. They select an entry and click 'Delete'. The application triggers a confirmation modal and executes an explicit SQL DELETE statement.
- **9. Generate Rankings (Read/Aggregate):** A user ranks regional performance parameters by choosing an evaluation metric from a dropdown. The backend runs a GROUP BY sorting calculation query to arrange data rows hierarchically.
- **10. View Dataset Information (Read):** A user accesses the 'About Data' page to review information integrity profiles explaining the FAOSTAT and World Bank schemas, row cardinalities, cleaning methodologies, and normalization rules.

8. Low-Fidelity UI Mockups



9. Project Work Distribution

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Relational Schema Design & Normalization Review	Database Creation, Ingestion Testing & DDL Setup, DCL	Front-to-Back Connected MVP (Retrieval & CRUD)	Transactions, Queries, Triggers, & Stored Procedures Integration
Implement front end UI and connect it with backend and database. Designs the interactive plots.	Creates and manages cloud database and connectivity	Creates backend functions that handle queries and functionality logic.	Data cleaning and validation from raw csv files.

Project work distribution subject to change